4.5 The Exponential Function

Definition 4.5.1 By

$$\exp(z) := e^z := \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad \cos(z) := \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}, \quad \sin(z) := \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}$$

we define three functions from $\mathbb{C}$ to $\mathbb{C}$, called the exponential, cosine, and sine.

Theorem 4.5.2

(1) The power series in Def. 4.5.1 all have radius of convergence $+\infty$.
(2) exp, sin, cos are real-valued on $\mathbb{R}$.
(3) $e^{z_1+z_2} = e^{z_1} \cdot e^{z_2}$.
(4) $e^z \neq 0$ for all $z \in \mathbb{C}$, and $e^{-z} = \dfrac{1}{e^z}$.
(5) $e^{iz} = \cos(z) + i\,\sin(z)$.
(6) cos is an even function, i.e. $\cos(-z) = \cos(z)$ for all $z \in \mathbb{C}$; sin is an odd function, i.e. $\sin(-z) = -\sin(z)$ for all $z \in \mathbb{C}$.
(7) For all $z \in \mathbb{C}$,

$$\cos(z) = \tfrac{1}{2}\big(e^{iz} + e^{-iz}\big), \qquad \sin(z) = \tfrac{1}{2i}\big(e^{iz} - e^{-iz}\big).$$

(8) For all $x \in \mathbb{R}$,

$$\cos(x) = \mathrm{Re}(e^{ix}), \qquad \sin(x) = \mathrm{Im}(e^{ix}).$$

(9) exp, cos, and sin are continuous.

Proof.

(1) Use the root test for the exponential series[1]: For a power series $\sum_{n=0}^{\infty} a_n z^n$ the radius of convergence is

$$\rho := \frac{1}{\limsup\limits_{n\to\infty} \sqrt[n]{|a_n|}} = \lim_{n\to\infty} \sqrt[n]{n!}$$

here with $a_n = 1/n!$. Using $n! \geq (n/2)^{n/2}$ we get

$$\rho \geq \lim_{n\to\infty} \sqrt{\tfrac{n}{2}} = +\infty.$$

Hence the radius of convergence is infinite.

(2) Clear.
(3) See Example 3.5.15.
(4) See Example 3.5.15.
(5) We have

$$e^{iz} = \sum_{n=0}^{\infty} \frac{(iz)^n}{n!} = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!} = \cos(z) + i\,\sin(z),$$

where we separated even and odd powers (the exponential series converges absolutely, so terms may be regrouped).

[1] For the exponential series one can also use the ratio test more directly; the root test applies analogously, and the same argument works for sin and cos.

(6) Trivial (consider the terms z^{2n} and z^{2n+1}).
(7) Using (5) and (6),

$$frac12\big(e^{iz}+e^{-iz}\big) \overset{(5)}{=} \tfrac{1}{2}\big(\cos(z)+i\sin(z)+\cos(-z)+i\sin(-z)\big) \overset{(6)}{=} \cos z,$$

and similarly

$$frac12i\big(e^{iz}-e^{-iz}\big) \overset{(5)}{=} \tfrac{1}{2i}\big(\cos(z)+i\sin(z)-\cos(-z)-i\sin(-z)\big) \overset{(6)}{=} \sin z.$$

(8) Follows from (2) and (5).
(9) Use (cf. Example 4.3.14): there exists $\rho > 0$ small enough such that

$$\left|\frac{e^z-1}{z}\right| < 2 \qquad \text{for all } |z| < \rho.$$

Let $z_0 \in \mathbb{C}$ and $\varepsilon > 0$. Choose $\delta := \min\{\rho, \frac{\varepsilon}{2|e^{z_0}|}\}$. Then for $|z-z_0| < \delta$ we have

$$|e^z - e^{z_0}| = |e^{z_0}(e^{z-z_0}-1)| \le |e^{z_0}| \left|\frac{e^{z-z_0}-1}{z-z_0}\right| |z-z_0| \le |e^{z_0}| \cdot 2\delta \le \varepsilon.$$

Hence e^z is continuous. For cos use $\cos(z) = \frac{1}{2}(e^{iz}+e^{-iz})$ and the fact that the compositions of continuous functions e^z and $z \mapsto \pm iz$ are continuous. The same argument yields continuity of sin.

■

Theorem 4.5.3 (Addition theorems) For all $z, w \in \mathbb{C}$ it holds:
(1) $\cos(z \pm w) = \cos z\ \cos w \mp \sin z\ \sin w$
$\sin(z \pm w) = \sin z\ \cos w \pm \cos z\ \sin w.$
(2) $\sin z - \sin w = 2\cos\big(\frac{z+w}{2}\big)\ \sin\big(\frac{z-w}{2}\big)$
$\cos z - \cos w = -2\sin\big(\frac{z+w}{2}\big)\ \sin\big(\frac{z-w}{2}\big).$
(3) $\cos^2 z + \sin^2 z = 1.$

Proof.
(1) Express $\cos, \sin$ via the exponential function:

$$\cos z\ \cos w \mp \sin z\ \sin w = \tfrac{1}{4}\Big\{(e^{iz}+e^{-iz})(e^{iw}+e^{-iw}) \mp \tfrac{1}{i}(e^{iz}-e^{-iz})\ \tfrac{1}{i}(e^{iw}-e^{-iw})\Big\} = \tfrac{1}{4}\big\{2e^{i(z+w)}+2e^{-i(z+w)}$$

The remaining identities in (1) are obtained analogously.
(2) Set $u = \frac{z+w}{2}$ and $v = \frac{z-w}{2}$. Using (1) we get

$$\sin(u+v) - \sin(u-v) = (\sin u\ \cos v + \cos u\ \sin v) - (\sin u\ \cos v - \cos u\ \sin v) = 2\cos u\ \sin v.$$

(3) Put $w = z$ in (1). Then

$$\cos^2 z + \sin^2 z = \cos(z-z) = \cos(0) = 1.$$

■

From Theorem 4.5.12 it follows: $e^z = e^{z+2k\pi i}$ for all $k \in \mathbb{Z}$. If a function $f : \mathbb{K} \to \mathbb{K}$ admits a $p \in \mathbb{K} \setminus \{0\}$ with $f(x+p) = f(x)$ for all $x \in \mathbb{K}$, we call f p-periodic. Hence the exponential function is $2\pi i$-periodic.

Theorem 4.5.13 For every $a \in \mathbb{R}$ the map

$$\mathrm{ixp} : [a, a + 2\pi[\to S^1$$

is bijective.

Proof.

(1) Injectivity:

$$\mathrm{ixp}(x) = \mathrm{ixp}(y) \iff e^{ix} = e^{iy} \iff e^{i(x-y)} = 1 \iff x - y = 2\pi k,\ k \in \mathbb{Z}.$$

For all $x, y \in [a, a + 2\pi[$ we have $|x - y| < 2\pi$, hence $x = y$.

(2) Surjectivity: Let $z \in S^1$. By Theorem 4.5.10 there exists $x \in \mathbb{R}$ with $\mathrm{ixp}(x) = z$. Write $x = a + 2k\pi + r$ with $r \in [0, 2\pi[$. Then

$$z = \mathrm{ixp}(x) = e^{i2k\pi}\, e^{i(a+r)} = \mathrm{ixp}(a + r),$$

so $a + r \in [a, a + 2\pi[$.

■

Theorem 4.5.14

(1) $\cos(z + 2k\pi) = \cos z$ and $\sin(z + 2k\pi) = \sin z$. Thus cosine and sine are 2π-periodic.
(2) $\cos z = 0 \iff z = \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$; and $\sin z = 0 \iff z = k\pi$, $k \in \mathbb{Z}$.
(3) $\sin(x) > 0$ for all $x \in]0, \pi[$, and $\sin(x)$ is strictly increasing on $[0, \frac{\pi}{2}]$.
(4) $\cos(z + \pi) = -\cos z$ and $\sin(z + \pi) = -\sin z$ for all $z \in \mathbb{C}$.
(5) $\sin\left(\frac{\pi}{2} - z\right) = \cos z$ and $\cos\left(\frac{\pi}{2} - z\right) = \sin z$ for all $z \in \mathbb{C}$.
(6) $\cos(\mathbb{R}) = \sin(\mathbb{R}) = [-1, 1]$.

Proof. Items (1), (2), (4), (5), (6) are left as exercises. For (3): From the proof of Theorem 4.5.10, part (3), we have $\sin(x) > 0$ for all $x \in]0, 1[$. From part (2) of that theorem it follows that $\sin(x) \neq 0$ for all $x \in]0, \pi[$. By continuity of sin (Intermediate Value Theorem, Theorem 4.4.7) we conclude $\sin(x) > 0$ on $]0, \pi[$.

For monotonicity: arguing as in the proof for sin, from $\cos(0) = 1$ and the alternating series estimates one obtains $\cos x > 0$ for all $x \in]0, \frac{\pi}{2}[$. If $0 \leq x < y \leq \frac{\pi}{2}$, then by Theorem 4.5.3 (addition theorems)

$$\sin y - \sin x = 2\cos\left(\frac{x + y}{2}\right) \sin\left(\frac{y - x}{2}\right) > 0,$$

which shows that sin is strictly increasing on $[0, \frac{\pi}{2}]$. ■

Let $z \in \mathbb{C}$, $z = x + iy$ with $x, y \in \mathbb{R}$. Then

$$e^z = e^{x+iy} = e^x \cdot e^{iy},$$

and the behavior of exp is determined by the behavior of $\exp : \mathbb{R} \to \mathbb{R}$ and $\exp : i\mathbb{R} \to \mathbb{R}$.

We first consider the real exponential function.

Theorem 4.5.4 R

(1) $0 < e^x < 1$ for all $x < 0$; $e^x > 1$ for all $x > 0$; and $e^0 = 1$.
(2) $\exp : \mathbb{R} \to \mathbb{R}_{>0}$ is strictly monotonically increasing.
(3) $\lim\limits_{x\to\infty} x^{-n}e^x = \infty$ for every $n \in \mathbb{N}$. Moreover, $\lim\limits_{x\to-\infty} e^x = 0$.

Proof.

(1) We have, for $x > 0$,

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!}x^n = 1 + \sum_{n=1}^{\infty} \frac{1}{n!}x^n > 1,$$

and for $x < 0$,

$$e^x = e^{-(-x)} = \frac{1}{e^{-x}} \in (0, 1).$$

(2) For $y > x$ it holds

$$e^y = e^x\, e^{y-x} > e^x,$$

because $y - x > 0$ and (1) implies $e^{y-x} > 1$.

(3) For $n \in \mathbb{N}$ and $x > 0$,

$$x^{-n}e^x \geq x^{-n}\,\frac{x^{n+1}}{(n+1)!} = \frac{x}{(n+1)!} \xrightarrow[x\to\infty]{} \infty.$$

Finally,

$$\lim_{x\to-\infty} e^x = \lim_{y=-x\to\infty} e^{-y} = \lim_{y\to\infty} \frac{1}{e^y} = 0.$$

■

Theorem 4.5.15 For every $a \in \mathbb{R}$ the map

$$\exp : \mathbb{R} + i\,[a, a + 2\pi[\;\to\; \mathbb{C} \setminus \{0\}$$

is bijective.

Proof.

(1) Injectivity: Let $z = x + iy$ and $w = u + iv$ with $x, y, u, v \in \mathbb{R}$. Then

$$e^z = e^w \iff e^{x+iy} = e^{u+iv} \iff e^{x-u} = e^{i(v-y)} \iff e^{x-u} = |e^{i(y-v)}| = 1 \implies x-u = 0$$

and hence $e^{i(y-v)} = 1 \implies y - v = 2\pi k$ with $k \in \mathbb{Z}$. Since $|y - v| < 2\pi$, we get $y = v$.

(2) Surjectivity: Let $z \in \mathbb{C} \setminus \{0\}$. Then $|z| > 0$ and we can write

$$z = |z| \cdot \frac{z}{|z|}, \qquad \frac{z}{|z|} \in S^1.$$

By Theorem 4.5.13 there exists a unique $y \in [a, a + 2\pi[$ with $e^{iy} = z/|z|$. Moreover, since $\exp : \mathbb{R} \to \mathbb{R}_{>0}$ is bijective, there is a unique $x \in \mathbb{R}$ with $e^x = |z|$. Therefore

$$e^{x+iy} = e^x\, e^{iy} = |z| \cdot \frac{z}{|z|} = z.$$

■

Definition 4.5.5 The inverse function of the real exponential function is called the (natural) logarithm and is denoted by ln.

$$\ln := (\exp|_{\mathbb{R}})^{-1} :]0, \infty[\to \mathbb{R}.$$

Theorem 4.5.6

(1) The logarithm is continuous and strictly increasing, with

$$\lim_{x\to\infty} \ln(x) = +\infty, \qquad \lim_{x\to 0^+} \ln(x) = -\infty.$$

(2) For all $x, y \in \mathbb{R}_{>0}$ it holds

$$\ln(x \cdot y) = \ln(x) + \ln(y), \qquad \ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y).$$

Proof.

(1) Follows from Theorem 4.4.8.

(2) Set $a := \ln(x)$ and $b := \ln(y)$. Then

$$x \cdot y = e^a \cdot e^b = e^{a+b} \implies \ln(x \cdot y) = a + b = \ln(x) + \ln(y).$$

The quotient rule follows analogously: $\ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$.

■

Extension of the power function Let $a > 0$. Up to now a^x was defined for $x = \frac{p}{q}$ with $p \in \mathbb{Z}$, $q \in \mathbb{N}_{\geq 1}$.

Goal: Define a^x for all $x \in \mathbb{R}$ so that the definition agrees with the previous one for $x \in \mathbb{Q}$.

Write $a = e^{\ln a}$. Then:

(1) For $x \in \mathbb{N}$ it holds

$$a^x = (e^{\ln a})^x = e^{\underbrace{\ln a + \cdots + \ln a}_{x\text{-times}}} = e^{x \ln a}.$$

(2) For $x \in \mathbb{N}$ it holds

$$a^{-x} = \frac{1}{a^x} = \frac{1}{e^{x \ln a}} = e^{-x \ln a}.$$

(3) For $x \in \mathbb{N}_{\geq 1}$ set $y := e^{\frac{1}{x} \ln a}$. Then

$$y^x = \left(e^{\frac{1}{x} \ln a}\right)^x = e^{\ln a} = a, \quad \text{i.e.} \quad a^{1/x} = \sqrt[x]{a} = e^{\frac{1}{x} \ln a}.$$

Let $x = \frac{p}{q}$ with $p \in \mathbb{Z}$, $q \in \mathbb{N}_{\geq 1}$. Then

$$a^x = \left(e^{\frac{1}{q} \ln a}\right)^p = e^{\frac{p}{q} \ln a}.$$

In summary, with the "previous" definition we have for all $x \in \mathbb{Q}$:

$$a^x = e^{x \ln a}.$$

Definition 4.5.7 For $a > 0$ and $x \in \mathbb{R}$, set

$$a^x := e^{x \ln a}.$$

Theorem 4.5.8 Let $a, b \in \mathbb{R}_{>0}$ and $x, y \in \mathbb{R}$. Then:

(1) $a^x a^y = a^{x+y}$, and $\dfrac{a^x}{a^y} = a^{x-y}$.
(2) $a^x b^x = (ab)^x$.
(3) $(a^x)^y = a^{xy}$.
(4) $\ln(a^x) = x \ln(a)$.

Proof. Exercise. ■

Theorem 4.5.9 For $\alpha \in \mathbb{R}_{>0}$ it holds:

$$\lim_{x\to\infty} x^{-\alpha} \ln(x) = 0, \qquad \lim_{x\to 0^+} x^{\alpha} \ln(x) = 0.$$

Proof. As $x \to \infty$ we set $y := \ln x \to \infty$. Then

$$\lim_{x\to\infty} x^{-\alpha} \ln(x) = \frac{1}{\alpha} \lim_{y\to\infty} \frac{\alpha y}{e^{\alpha y}} = \frac{1}{\alpha} \lim_{z\to\infty} \frac{z}{e^z} = 0$$

by Theorem 4.5.4(3). Similarly,

$$\lim_{x\to 0^+} x^{\alpha} \ln(x) = \lim_{x\to 0^+} \left(\frac{1}{x}\right)^{-\alpha} \ln\left(\frac{1}{x}\right) = -\lim_{y\to\infty} y^{-\alpha} \ln y = 0.$$

■

Next we investigate the behavior of the exponential function on $i\mathbb{R} \subset \mathbb{C}$ and, for brevity, introduce the notation

$$\operatorname{ixp} : \mathbb{R} \to \mathbb{C}, \qquad \operatorname{ixp}(x) := e^{ix}.$$

Theorem 4.5.10 $\operatorname{ixp}(\mathbb{R}) = S^1 := \{z \in \mathbb{C} : |z| = 1\}$.

Proof.

(1) The inclusion $\operatorname{ixp}(\mathbb{R}) \subset S^1$ follows from

$$|\operatorname{ixp}(x)| = |e^{ix}| = |\cos x + i \sin x| = \sqrt{\cos^2 x + \sin^2 x} = 1.$$

(2) Let $I := (-1, 1)$. We show: the function $\sin : I \to \mathbb{R}$ is strictly increasing.
Using the power series of sin we obtain for $u, v \in I$:

$$\sin(u) - \sin(v) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (u^{2n+1} - v^{2n+1}) = \sum_{n=0}^{\infty} (u-v) \frac{(-1)^n}{(2n+1)!} \sum_{j=0}^{2n} u^j v^{2n-j} =: (u-v)\,(1+R(u,v)) \tag{$*$}$$

For $u, v \in I$ we have the bound

$$|R(u,v)| \le \sum_{n=1}^{\infty} \frac{1}{(2n+1)!(2n+1)} \le \sum_{n=1}^{\infty} \frac{1}{(2n)!} < \sum_{n=0}^{\infty} \frac{1}{2^{2n+1}} = \tfrac{1}{2} \sum_{n=0}^{\infty} \left(\tfrac{1}{4}\right)^n = \tfrac{1}{2} \frac{1}{1 - 1/4} = \tfrac{2}{3}.$$

Hence $|1 + R(u,v)| > 1/3$, and by $(*)$ the sign of $\sin(u) - \sin(v)$ equals the sign of $u - v$. Thus sin is strictly increasing on I.

(3) We now show that from (2) it follows that I is mapped injectively by ixp onto an arc $\tilde{y} \subset S^1$ that passes through 1 from below to above.
Since sin is strictly increasing and continuous on I, its image $J := \sin(I)$ is an open interval. Clearly $\mathrm{ixp}(0) = e^0 = 1 \in \tilde{y}$. Because $\sin(-x) = -\sin(x)$ and I is symmetric, we have

$$J = (-\alpha, \alpha) \quad \text{for some } \alpha \in (0, 1).$$

For $0 < x < 1$, the series of sin and cos are alternating; as in the proof of Theorem 3.4.8 (Leibniz criterion) this yields

$$\cos(x) > 1 - \frac{x^2}{2} > 0, \qquad 0 < x\Big(1 - \frac{x^2}{6}\Big) < \sin(x) < x, \quad x \in (0, 1). \qquad (**)$$

Hence $\cos x > 0$ and $\sin x$ changes sign only at 0, so ixp maps I injectively onto an open arc of S^1 going from below to above through 1.

(4) We show that I can be shrunk to a symmetric interval $[-\lambda, \lambda]$ such that $\mathrm{ixp} : [-\lambda, \lambda]$ traces exactly a quarter-circle arc with apex at 1.
From $(**)$ we get

$$\sin\Big(\tfrac{1}{\sqrt{2}}\Big) < \tfrac{1}{\sqrt{2}}, \qquad \sin\Big(\tfrac{4}{5}\Big) > \tfrac{4}{5}\Big(1 - \tfrac{16}{6\cdot 25}\Big) > \tfrac{4}{5}\Big(1 - \tfrac{16}{144}\Big) = \tfrac{32}{45} = \frac{\sqrt{1024}}{\sqrt{2025}} > \sqrt{\tfrac{1}{2}}.$$

By monotonicity and continuity of sin on I there exists a unique $\lambda \in (1/\sqrt{2},\, 4/5)$ with $\sin\lambda = 1/\sqrt{2}$. Thus $\mathrm{ixp}([-\lambda, \lambda])$ is precisely the quarter-arc of S^1 centered at 1.

(5) Continuity of sin ensures the existence of such a $\lambda \in (1/\sqrt{2},\, 4/5)$ (intermediate value theorem) with $\sin\lambda = 1/\sqrt{2}$. From $(**)$ we also get

$$\cos\lambda = +\sqrt{1 - \sin^2\lambda} = \sqrt{\tfrac{1}{2}},$$

hence

$$e^{\pm i\lambda} = \cos\lambda \pm i\sin\lambda = \tfrac{1}{\sqrt{2}}(1 \pm i).$$

(6) We claim: ixp maps the interval $[0, 8\lambda]$ bijectively onto S^1 (up to the identification $\mathrm{ixp}(0) = \mathrm{ixp}(8\lambda) = 1$).
Preparation:

$$e^{i\,2\lambda} = (e^{i\lambda})^2 = \Big(\tfrac{1}{\sqrt{2}}(1 + i)\Big)^2 = \tfrac{1}{2}(1 - 1 + 2i) = i, \quad e^{i\,4\lambda} = -1, \quad e^{i\,8\lambda} = 1.$$

We already showed that $\mathrm{ixp} : [-\lambda, \lambda] \to y$ (the quarter arc of S^1 with apex 1) is bijective. For $t \in [\lambda, 3\lambda]$ we compute

$$\mathrm{ixp}(t + 2\lambda) = e^{i(t+2\lambda)} = e^{it}\, e^{i\,2\lambda} = i\ \mathrm{ixp}(t) =: T(\mathrm{ixp}(t)),$$

where $T(z) := iz$ is the rotation of the complex plane by $90°$ counterclockwise about the origin. Hence

$$\mathrm{ixp}([\alpha + 2\lambda, \beta + 2\lambda]) = T\big(\mathrm{ixp}([\alpha, \beta])\big).$$

Iterating this 90° rotation shows that the intervals

$$[-\lambda, \lambda], \quad [\lambda, 3\lambda], \quad [3\lambda, 5\lambda], \quad [5\lambda, 7\lambda], \quad [7\lambda, 9\lambda]$$

are mapped bijectively onto the quarter arcs of S^1 with apices $1, i, -1, -i, 1$, respectively. In particular, ixp maps $[0, 8\lambda]$ bijectively onto S^1 (with the exception that 0 and 8λ both map to 1).

Theorem 4.5.11 The set

$$M := \{\, x \in \mathbb{R}_{>0} : \operatorname{ixp}(x) = 1 \,\}$$

has a positive minimum. We define

$$\pi := \tfrac{1}{2} \min M.$$

Proof. In the proof of Theorem 4.5.10 (part (5)) we showed that the map

$$\operatorname{ixp} :]0, 8\lambda] \to S^1$$

is bijective. Hence there exists an inverse

$$\phi := (\operatorname{ixp}|_{]0,8\lambda]})^{-1} : S^1 \to]0, 8\lambda].$$

It follows that $x := \phi(1)$ lies in $]0, 8\lambda]$ and is uniquely determined. Consequently there is no smaller $y \in \mathbb{R}_{>0}$ with $\operatorname{ixp}(y) = 1$. We denote this value by

$$x = 2\pi.$$

Since $1 = e^{i2\pi} = \operatorname{ixp}(8\lambda)$ (by part (5) of the proof of Theorem 4.5.10) and $8\lambda \in]0, 8\lambda]$, we have $\phi(1) = 8\lambda$ and therefore

$$\pi = 4\lambda, \qquad \lambda = \tfrac{\pi}{4}.$$

From this we obtain the familiar values

$$e^{i\pi/2} = i, \qquad e^{i\pi} = -1, \qquad e^{2i\pi} = 1.$$

■

Theorem 4.5.12

(1) $e^z = 1$ if and only if $z \in 2\pi i\,\mathbb{Z}$, i.e., $z = 2\pi i\, k$ for $k \in \mathbb{Z}$.

(2) $e^z = -1$ if and only if $z \in \pi i + 2\pi i\,\mathbb{Z}$, i.e., $z = (2k+1)\pi i$ for $k \in \mathbb{Z}$.

Proof.

(1) "$\Leftarrow$": $e^{2\pi i k} = (e^{2\pi i})^k = 1^k = 1$.

"$\Rightarrow$": Let $z = x + iy$, $x, y \in \mathbb{R}$, and suppose $e^z = 1$. Then

$$1 = |e^z| = |e^{x+iy}| = |e^x|\,|e^{iy}| = e^x \implies x = 0.$$

Write $y = 2k\pi + r$ with $r \in [0, 2\pi[$. Then

$$1 = e^{i2\pi k} e^{ir} \implies e^{ir} = 1 \implies r = 0.$$

Hence $y = 2k\pi$ and $z = 2\pi i\, k$.

(2) "$\Leftarrow$": $e^{(2k+1)\pi i} = e^{\pi i} e^{2k\pi i} = (-1) \cdot 1 = -1$.
"$\Rightarrow$": If $-1 = e^z$, then $1 = e^{2z}$, and by (1) we get $2z = 2k\pi i$, hence $z = k\pi i$. In particular, when k is even, $z = k\pi i$ does not solve $e^z = -1$; therefore k must be odd and $z = (2k+1)\pi i$.

■

From this we obtain the polar-coordinate representation of complex numbers.

Theorem 4.5.16 For every $z \in \mathbb{C} \setminus \{0\}$ there exists a unique $\alpha \in [0, 2\pi[$ such that

$$z = |z|\, e^{i\alpha}.$$

We call α the argument of z; write $\alpha := \arg(z)$.

Theorem 4.5.17 For every $a \in \mathbb{C} \setminus \{0\}$ the equation $z^n = a$ has exactly n distinct solutions:

$$z_k = |a|^{1/n} \exp\left(\frac{i(\arg(a) + 2\pi k)}{n} \right), \qquad k = 0, 1, \ldots, n-1.$$

Proof. Let z be a solution of $z^n = a$ (clearly $z \neq 0$). Use polar coordinates

$$a = |a| e^{i \arg(a)}, \qquad z = r\, e^{i\phi}, \quad r \in \mathbb{R}_{>0},\ \phi \in [0, 2\pi[.$$

Then

$$z^n = r^n e^{in\phi} = |a| e^{i \arg(a)} \iff \begin{cases} r^n = |a|, \\ e^{in\phi} = e^{i \arg(a)}. \end{cases}$$

The first condition gives $r = |a|^{1/n}$. The second says $e^{i(n\phi - \arg(a))} = 1$, hence $n\phi - \arg(a) = 2\pi k$ and

$$\phi = \tfrac{1}{n}\big(\arg(a) + 2\pi k\big), \qquad k \in \mathbb{Z}.$$

Distinct solutions arise exactly for $k = 0, 1, \ldots, n-1$. ■